



Scanning Networks Assignment (Core)

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Coding Dojo Cybersecurity Boot Camp
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Assignment

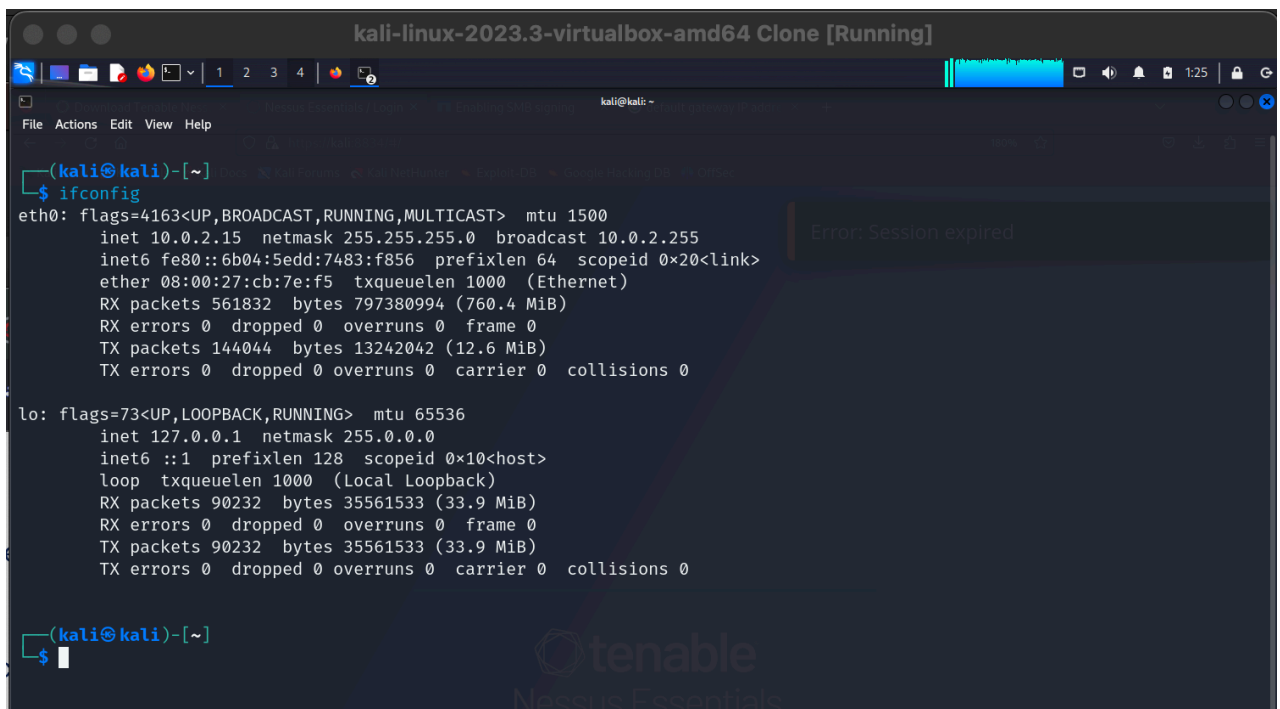
- Perform a network scan with Nmap.

Assignment Submission

Upload a single document including the following:

1. Screenshots of your scan results.
2. Summary of the results in your own words.
3. For each IP address on the scan, approximate what services are running on each device based on the ports open.

To find my default gateway and subnet mask so that I know what range of IP addresses to scan, I use the '*ifconfig*' command. Screenshot is below:



```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
    inet6 fe80::6b04:5edd:7483:f856  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:cb:7e:f5  txqueuelen 1000  (Ethernet)
    RX packets 561832  bytes 797380994 (760.4 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 144044  bytes 13242042 (12.6 MiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 90232  bytes 35561533 (33.9 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 90232  bytes 35561533 (33.9 MiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

(kali@kali)-[~]
$
```

I use the '**nmap**' command using **sudo** privileges to initiate a network scan.
Please find below the screenshots of the scan results:

```
(kali@kali)-[~]
$ sudo nmap 10.0.2.1-254
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-11-15 01:27 EST
Nmap scan report for 10.0.2.1
Host is up (0.00036s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)

Nmap scan report for 10.0.2.2
Host is up (0.00038s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
5000/tcp  open  upnp
7000/tcp  open  afs3-fileserver
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)

Nmap scan report for 10.0.2.3
Host is up (0.00035s latency).
All 1000 scanned ports on 10.0.2.3 are in ignored states.
Not shown: 1000 filtered tcp ports (proto-unreach)
MAC Address: 08:00:27:6F:3D:44 (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.15
Host is up (0.0000030s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

Nmap done: 254 IP addresses (4 hosts up) scanned in 3.38 seconds

(kali@kali)-[~]
$
```

Summary of the results in your own words.

For each of the IP addresses specified in the network range, the scan results reveal the status of the host and its latency, the ports that are opened, the state of each of the opened port and the likely service running on them. Of the entire 254 IP addresses scanned, only 4 hosts were found to be active.

Here is a breakdown of the findings for each of the active host:

- **10.0.2.1:**
 - Host is up with low latency (0.00036s).
- **10.0.2.2:**
 - Host is up with low latency (0.00038s).
- **10.0.2.3:**
 - Host is up with low latency (0.00035s).
- **10.0.2.15:**
 - Host is up with extremely low latency (0.0000030s).

For each IP address on the scan, approximate what services are running on each device based on the ports open.

1. **10.0.2.1:**

- Running a domain service on port 53/tcp

2. **10.0.2.2:**

- Running UPnP service on port 5000/tcp.
- Running AFS3-fileserver service on port 7000/tcp.

3. **10.0.2.3:**

- All 1000 scanned ports are in ignored states.
- 1000 filtered tcp ports (proto-unreach).

4. **10.0.2.15:**

- Running NetBIOS-SSN service on port 139/tcp.
- Running Microsoft-DS service on port 445/tcp.

In summary, based on the open ports:

- **10.0.2.1** is likely a DNS server.
- **10.0.2.2** is hosting UPnP and AFS3-fileserver services.
- **10.0.2.3** has all ports in ignored states and 1000 filtered ports.
- **10.0.2.15** is likely running NetBIOS-SSN and Microsoft-DS services, typical of Windows file and printer sharing.